

CSSCM 2026 COHORT

EXERCISE 3

Five-Species Spectrophotometric Analysis with NumPy

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Scenario

A laboratory is analysing a mixture containing five absorbing chemical species **A**, **B**, **C**, **D** and **E**. The mixture is measured at five wavelengths. After applying the Beer-Lambert relationship, the absorbance contribution at each wavelength can be represented as a linear combination of the five unknown concentrations.

Assume a path length of 1 cm. The numerical coefficients below represent the effective response of each species at each wavelength.

$$4C_A + C_B + 2C_C + C_D + 3C_E = 9.8$$

$$C_A + 5C_B + C_C + 2C_D + C_E = 11.3$$

$$2C_A + C_B + 4C_C + C_D + 2C_E = 8.5$$

$$C_A + 2C_B + C_C + 5C_D + C_E = 12.2$$

$$3C_A + C_B + 2C_C + C_D + 4C_E = 9.9$$

Matrix Form

Represent the system as $EC = A$:

$$E = \begin{bmatrix} 4 & 1 & 2 & 1 & 3 \\ 1 & 5 & 1 & 2 & 1 \\ 2 & 1 & 4 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 \\ 3 & 1 & 2 & 1 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} C_A \\ C_B \\ C_C \\ C_D \\ C_E \end{bmatrix}, \quad A = \begin{bmatrix} 9.8 \\ 11.3 \\ 8.5 \\ 12.2 \\ 9.9 \end{bmatrix}$$

Part A - Build and Inspect the Arrays

1. Import NumPy as np.
2. Create the coefficient matrix E as a 5 x 5 NumPy array.
3. Create the absorbance vector A.
4. Print E.shape, E.ndim, E.size, and A.shape.
5. Confirm from the output that there are five equations and five unknown concentrations.

Part B - Solve the 5 x 5 Chemical System

6. Solve $EC = A$ using `np.linalg.solve(E, A)`.
7. Store the result in an array called C.
8. Print the complete concentration vector.
9. Print C_A , C_B , C_C , C_D and C_E separately to four decimal places.

Part C - Verify the Numerical Solution

10. Recalculate the absorbance vector using `calculated_A = E @ C`.
11. Print both `calculated_A` and the original A.
12. Use `np.allclose(calculated_A, A)` to test whether the solution reproduces the five measurements.

13. Calculate the residual vector:

$$\text{residual} = A - \text{calculated_A}$$

For a correctly solved square system, explain why these residuals should be zero or extremely close to zero.

Part D - Challenge: An Independent Sixth Wavelength

The chemist now measures the same mixture at a **sixth wavelength**. The response coefficients at this wavelength are:

```
validation_coefficients = [2, 3, 1, 2, 4]
```

The experimentally measured absorbance at the sixth wavelength is:

$$A_{6, \text{measured}} = 12.55$$

14. Use the concentration vector obtained in Part B to predict the sixth-wavelength absorbance:

$$A_{6, \text{predicted}} = \text{validation_coefficients} @ C$$

15. Calculate the absolute residual:

$$|A_{6, \text{measured}} - A_{6, \text{predicted}}|$$

16. Calculate the percentage error relative to the measured value:

$$\% \text{ error} = |A_{6, \text{measured}} - A_{6, \text{predicted}}| / A_{6, \text{measured}} \times 100$$

17. The laboratory accepts the model if the validation error is less than 2%. Use an if/else statement to print either `Validation passed` or `Validation failed`.

Part E - Advanced Challenge: Use All Six Measurements

The first five equations form a square 5 x 5 system. Adding the sixth wavelength produces a **6 x 5 system**: six measurements but only five unknown concentrations. Such a system is overdetermined and should not be passed to `np.linalg.solve()`.

18. Create a new 6 x 5 matrix by adding the sixth coefficient row to E.

19. Create a new six-element measurement vector by adding 12.55 to the original absorbance vector.

20. Solve the overdetermined system using least squares:

```
C_ls, residuals, rank, singular_values = np.linalg.lstsq(
    E6, A6, rcond=None
)
```

21. Compare the original 5 x 5 concentration vector C with the least-squares vector C_ls.

22. Calculate all six predicted absorbances using E6 @ C_ls.

23. Calculate the six residuals and identify the measurement with the largest absolute residual using `np.argmax(np.abs(...))`.

24. In one or two sentences, explain why a least-squares solution may be useful when experimental measurements contain small amounts of noise.

Useful NumPy Hints

```
# Solve a square system
C = np.linalg.solve(E, A)

# Matrix-vector multiplication
predicted = E @ C

# Numerical comparison
np.allclose(predicted, A)

# Join an additional row to a 2D array
E6 = np.vstack((E, new_row))

# Join an additional value to a 1D array
A6 = np.append(A, new_value)

# Least-squares solution for an overdetermined system
```

```

C_ls, residuals, rank, singular_values = np.linalg.lstsq(
    E6, A6, rcond=None
)

# Position of largest absolute residual
index = np.argmax(np.abs(residual_vector))

```

Note: These are syntax hints only. They do not contain the numerical solution.

Expected Terminal Report

```

=====
EXERCISE 3 - FIVE-SPECIES ANALYSIS
=====

Matrix shape: _____
Matrix dimensions: _____
Number of matrix elements: _____

Concentration vector:
[...]

Species A = _____
Species B = _____
Species C = _____
Species D = _____
Species E = _____

Original absorbances:
[...]

Calculated absorbances:
[...]

Square-system verification: _____

-----
SIXTH-WAVELENGTH VALIDATION
-----
Predicted absorbance: _____
Measured absorbance : 12.55
Absolute residual   : _____
Percentage error    : _____ %
Validation result    : _____

-----
SIX-MEASUREMENT LEAST-SQUARES CHALLENGE
-----
Least-squares concentrations:
[...]

Six predicted absorbances:
[...]

Residual vector:
[...]

Largest residual occurs at measurement: _____
=====

```

Interpretation Questions

- Why is the original coefficient matrix 5 x 5?
- What does each row represent chemically?
- What does each column represent chemically?
- Why can `np.linalg.solve()` be used for the original system but not directly for the 6 x 5 system?
- What information does a residual provide about a numerical model and an experimental measurement?
- If one wavelength has a much larger residual than all the others, what experimental issues might you investigate?

Submission

Save the program as `exercise3_five_species.py` and run it in Ubuntu with:

```
python3 exercise3_five_species.py
```

Submit the Python script and a screenshot of the completed terminal output. All numerical results must be produced by the program.